

SHOAIB ASGHAR

Software Engineer | Full-Stack & AI | MERN · TypeScript · Python · Automation

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PROFESSIONAL SUMMARY

Software Engineer with experience building MERN stack applications, AI-integrated systems, and data analytics tools across web and mobile platforms. Proficient in React, Node.js, TypeScript, Python, and MongoDB, with applied depth in LLM integration and RAG architecture, geospatial web development, and asynchronous backend systems. Experienced in REST API design, statistical data visualization, and end-to-end testing. Built multiple projects spanning civic technology, e-commerce, analytics, and automation, with a focus on production-grade architectural patterns. Proficient in Agile methodologies, software quality assurance, and SDLC, with grounding in software project management. Worked on multiple Android apps as well. Ranked 128th of 33,000+ candidates, top 0.4%, 99.6th percentile, in the National Skill Competency Test (NSCT), organized by HEC and P@SHA, 2026.

EDUCATION

Bachelor of Science in Software Engineering

HITEC University, Taxila | 2022 – 2026 | CGPA: 3.91 /4.0

PROJECTS

Jamhur: Smart Civic Action Platform for Local Issues (FYP)

MERN · TypeScript · BullMQ · Vercel AI SDK · Leaflet · RTK Query · Tailwind CSS · Shadcn UI · Cloudinary · Vitest · Cypress

- Full-Stack Civic Platform & Geospatial Mapping: Web App for reporting, discussing, and resolving local issues and organizing and coordinating community initiatives through a structured process. Including Leaflet for interactive GIS local areas mapping, MongoDB 2dsphere indexes for geographic queries, and the Nominatim API to map GPS coordinates to Pakistani areas.
- AI Copilot & RAG Pipeline: Integrated a streaming AI copilot (Vercel AI SDK & Google Gemini) utilizing a Retrieval-Augmented Generation (RAG) pipeline to query local civic laws and past community resolutions, implementing geographic pre-filtering and token budgeting to keep context highly relevant and cost-effective.
- Knowledge Base & Feedback Loop: Designed a wiki-style civic knowledge base with version control and community voting. Automated content embedding ingestion using BullMQ, and created an AI feedback loop that summarizes resolved community issues and indexes them as precedents for future recommendations, creating a continuously improving pipeline
- Asynchronous Content Moderation: Built a background moderation pipeline using BullMQ and Redis to process multilingual content (English, Urdu, and Roman Urdu), featuring instant post visibility, a fail-open design to ensure uptime, and custom rules that permit constructive civic complaints while filtering abuse.

AI-Powered Dispute Resolution Engine

Node.js · PostgreSQL · MongoDB · Google Gemini · Cloud Vision API · Sightengine

- Hybrid AI Orchestration Pipeline: Engineered a 5-stage dispute resolution engine combining deterministic Node.js logic with LLM synthesis, strictly preventing AI hallucinations and prompt injection.
- Image Forensics & Red Teaming: Integrated Vision and Sightengine APIs to computationally scan evidence for deepfakes and EXIF integrity. Developed a dual-LLM architecture where a "Critique AI" audits the primary model for logical leaps.

- **Deterministic Fact-Checking:** Built a pre-processing stage using PostgreSQL and Regex to verify raw transaction facts before LLM execution, generating a continuous feedback loop dataset from human overrides.

Docent: AI Documentation Assistant

[React](#) · [Node.js](#) · [FastAPI](#) · [Qdrant](#) · [BAAI Reranker](#) · [DeepEval & Ragas](#) · [Playwright](#) · [Docker](#)

- **Hybrid RAG Architecture:** Engineered an AI documentation assistant using an API Gateway pattern, combining Qdrant vector search with cross-encoder semantic reranking to eliminate LLM hallucinations.
- **Security & Guardrails:** Designed a hardened Node.js API Gateway enforcing strict payload caps, cascading timeouts, and prompt injection defenses with least-privilege containerized services.
- **Automated Evaluation & CI/CD:** Built a 4-tier testing strategy (Playwright, Pytest) and automated GitHub Actions workflows integrating continuous security scanning and compliance checks.

Ezitech iPortal

[React](#) · [Material-UI \(MUI\)](#) · [Framer Motion](#) · [Node.js](#) · [Express](#) · [MySQL](#) · [GitHub](#)

A multi-role management system built using React, Material-UI, Node.js, and MySQL. It coordinates organizational workflows between interns, supervisors, and universities by managing onboarding (remote/onsite), task tracking, leave requests, and financial withdrawals through interactive dashboards and automated background jobs.

Film Analytics App (IMDb user data)

[React](#) · [TypeScript](#) · [ApexCharts](#) · [Zustand](#) · [Zod](#)

Local-first analytics dashboard for visualizing personal IMDb viewing history and film preferences. Processes imported ratings data to generate statistical insights across genres, directors, temporal patterns, rating psychology, and taste profiling. Runs entirely in the browser with no server dependency.

Mughal Electronics Store App

[MERN](#) · [Tailwind](#) · [Jest](#) · [Cypress](#), [Store AI Chatbot](#)

A full-stack e-commerce platform for electronics retail on the MERN stack, with JWT/bcrypt auth and a full test suite (Jest, Supertest, Cypress E2E). The store chatbot pulls the live product catalog and FAQs into each request before answering

ReelVision: AI Batch URL Extractor

[Python](#) · [ffmpeg](#) · [gallery-dl](#) · [Gemini Vision](#)

Automated AI pipeline that processes batches of Instagram reels to extract website URLs displayed on-screen, using a vision LLM to locate and read browser address bars directly from video frames. Features ephemeral disk management, targeted frame sampling, idempotent and resumable execution, and structured Markdown output.

Multilingual NLP Civic Complaint Processor

[Python](#) · [NLP](#) · [Transformer Models](#)

Developed an NLP pipeline for processing Roman Urdu civic complaints. Utilized pre-trained Transformer models for Named Entity Recognition (NER) and Intent Classification, implementing BIO tagging and inference workflows to extract structured data such as locations and severity.

FXStore

[Java](#) · [JavaFX](#) · [MVC](#) · [MySQL](#) · [JDBC](#) · [Gradle](#)

A desktop e-commerce app built with Java 21 and JavaFX, structured around clean MVC + DAO separation - FXML views, service-layer business logic, a pure-JDBC data layer. Full shopping flow: auth, product catalog, cart, checkout with receipt, order history, backed by MySQL.

Academic Projects, Portals, Management Systems

Multiple academic and hobby projects including but not limited to:

- Route Navigation System — Java · Dijkstra's Algorithm
- Hostel Management System — Web App · NodeJS · MongoDB · JavaScript

WORK EXPERIENCE

Full Stack Developer — Ezitech Institute

July 2025 – October 2025

- Maintained, added features, and resolved legacy code issues in the organization's central portal (iPortal)
- Built and updated UI using React, Material UI, Framer Motion on the admin dashboard to manage company workflow, intern accounts, assignments, and payouts.
- Used Node.js/Express APIs and MySQL queries to handle incoming data from interns, supervisors, and partner universities.
- Wrote background scripts to handle recurring tasks like email verifications and status updates.
- Managed code changes and collaborated with the team using GitHub.

General Secretary and Full Stack Developer — InnoNerds

October 2024 – present

- Managed end-to-end organizational operations and documentation as General Secretary, while architecting and deploying full-stack web applications for events and competitions.
- Streamlined internal workflows and organized events for tech initiatives, including Code Battle Season 2 and various specialized technical workshops.
- Mentored junior developers on full-stack development and learning and adopting latest tech trends

ACHIEVEMENTS

Ranked in the top 0.4% candidates, with 128th position out of 33,000+ candidates, achieving the 99.6 percentile in National Skill Competency Test (NSCT) — organized by HEC and P@SHA in 2026.

Reference: <https://www.hec.gov.pk/english/scholarshipsgrants/Pages/Sites/NSCT-Students.aspx>

CERTIFICATIONS

Mobile Application Developer — NAVTTC

Responsive Web Design, JavaScript Algorithms and Data Structures — FreeCodeCamp

The Bits and Bytes of Computer Networking — Google

DeepLearning.AI: Generative AI for Software Development Specialization — DeepLearning.AI

Introduction to Front-End Development, Programming with JavaScript, Version Control, React — Meta (Coursera)

Learn to work with APIs Specialization — Scrimba

SKILLS AND TECHNOLOGIES

Languages	JavaScript, TypeScript, Python, Java, C++, SQL, Bash
Frontend	React, Redux & RTK Query, TailwindCSS, Shadcn UI, MUI, Framer Motion
Backend	Node.js, Express.js, FastAPI, BullMQ, JWT, bcrypt, Multer
Databases	MongoDB (Mongoose, Atlas Vector Search), PostgreSQL, MySQL, Redis, Qdrant
AI & LLM	RAG Architecture, Google Gemini, Vercel AI SDK, Embeddings, NLP
Testing	Vitest, Cypress, Playwright, Jest, DeepEval, Ragas
Tools/Platforms	Git, GitHub Actions, Docker, Postman, Figma, Cloudinary, ffmpeg, Android Studio
Methodology	Agile, SDLC, Software Quality Assurance, Software Project Management